

Nguyen Luong Gia Bao

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PARTICULARS

EDUCATION

University of Economics and Law – Vietnam National University 2023 – 2027
Bachelor of Financial Technology
GPA: 3.43/4.0 (8.22/10)

RESEARCH INTERESTS

- **Financial Behavior & Empirical Finance:** causal inference, econometric methods, and event-study analysis
- **Financial Engineering & Applied AI:** derivative pricing, risk management, deep learning, reinforcement learning, and graph neural networks
- **Algorithmic Game Theory & Market Design:** equilibrium computation in continuous double auctions, mechanism design, and multi-agent learning with realistic market frictions

DISSERTATION

Title: “State-Dependent Graph Portfolio Optimization using Selective State Space Models and Regime-Adaptive Graph Learning”

Advisor: MSc. Ngo Phu Thanh

My dissertation examines regime-dependent financial networks and their role in portfolio optimization. The study develops a selective state space framework to model asset-return dynamics across changing market regimes, while a regime-adaptive graph learning mechanism captures time-varying cross-asset dependencies. The resulting state-dependent graph representation is incorporated into a portfolio construction pipeline that adapts to market conditions and seeks stronger risk-adjusted performance during periods of heightened volatility and regime transition.

RESEARCH EXPERIENCE

National Yang Ming Chiao Tung University January 2027 – June 2027 (Accepted)
Hsinchu, Taiwan

- Research Intern at the Economics and Computing Laboratory – PI: Prof. Dr. Po-An Chen, Director of Institute of Information Management.
- Investigating explainable AI methods for transparent and interpretable decision systems in economics and finance.
- Exploring algorithmic game theory and theoretical machine learning for strategic decision-making and multi-agent optimization.
- Assisting an ongoing 3-year NSTC research project (2024–2027) on convergences in multiagent online/reinforcement learning and robust market making; contributing to the extension and integration of results across related ongoing manuscripts.

INDUSTRY EXPERIENCE

Miquant – Quantitative Research Analyst October 2025 – Present
Ho Chi Minh City, Vietnam

- Conducted quantitative research and analysis for financial markets using advanced statistical methods
- Developed and implemented mathematical models and algorithms for optimized trading strategies
- Analyzed complex market data and designed comprehensive risk management frameworks

Lumination – Data Analyst August 2025 – October 2025
Ho Chi Minh City, Vietnam

- Built time-series data pipelines for forecasting and anomaly detection across multiple business domains
- Designed interactive dashboards with automated reporting, reducing manual reporting effort by streamlining data workflows

- Applied statistical modeling and feature engineering techniques to extract actionable patterns from large, complex datasets

Vikki Digital Bank – Risk Consultant
Ho Chi Minh City, Vietnam

February 2025 – May 2025

- Developed risk scoring models using statistical and machine learning approaches to quantify credit and operational risk
- Conducted data-driven risk analysis under regulatory constraints, identifying vulnerabilities in digital banking products
- Designed risk management frameworks and delivered strategic recommendations to senior management based on quantitative findings

PUBLICATIONS AND PRESENTATIONS

UNDER REVIEW

1. Ha Quoc Hung, Ngo Phu Thanh, **Nguyen Luong Gia Bao** (2026), “MambaLOB: Efficient and Uncertainty-Aware Limit Order Book Forecasting for Cryptocurrency Markets,” *International Journal of Information Management Data Insights* (Scopus Q1).
2. **Nguyen Luong Gia Bao**, Ngo Phu Thanh (2026), “State-Dependent Graph Portfolio Optimization using Selective State Space Models and Regime-Adaptive Graph Learning,” *Int. J. of Computational Economics and Econometrics* (ESCI Q4, ABDC C).
3. Ha Quoc Hung, Ngo Phu Thanh, **Nguyen Luong Gia Bao** (2026), “Semantic Stacking for Equity Ranking: Cross-Sectional Temporal Encoders with Learning-to-Rank,” *Quantitative Finance* (SSCI Q2, IF: 1.4).
4. Ha Quoc Hung, Ngo Phu Thanh, **Nguyen Luong Gia Bao** (2026), “Period-Aware Analog Retrieval for Financial Risk Forecasting,” *Computational Economics* (SSCI Q2, IF: 2.1).
5. **Nguyen Luong Gia Bao**, Ngo Phu Thanh (2026), “World Uncertainty, Cost of Debt, and Firm Financial Stability: Nonlinear and Heterogeneous Evidence,” *Asian Development Review* (ESCI Q2, IF: 1.4).
6. Tran Phuoc, Ngo Phu Thanh, **Nguyen Luong Gia Bao** (2026), “World Uncertainty, Cost of Debt, and Corporate Financial Stability: Panel and Machine-Learning Evidence,” *Research in Economics* (ESCI Q3, IF: 1.1).
7. **Nguyen Luong Gia Bao**, Nguyen Phat Dat, Ngo Phu Thanh (2026), “Market Breadth as a State-Dependent Return Signal: A Double Machine Learning Approach,” *Discover Sustainability* (ESCI Q1, IF: 5.2).
8. **Nguyen Luong Gia Bao**, Ngo Phu Thanh (2026), “Economic Policy Uncertainty, Corporate Cash Buffers, and Firm Financial Stability: Double Machine Learning Evidence from Southeast Asia,” *Cogent Economics & Finance* (ESCI Q2, IF: 1.7, ABDC B).
9. **Nguyen Luong Gia Bao**, Ngo Phu Thanh (2026), “Policy Uncertainty, Cash Buffers, and Firm Financial Stability: Evidence from Nonlinear and Machine-Learning Approaches,” *Macroeconomics and Finance in Emerging Market Economies* (ESCI Q3, IF: 1.1, ABDC C).
10. Tam Phan Huy, Vu Tran Huu Anh, **Nguyen Luong Gia Bao**, Tien Vo Quoc, Thu Truong Ngoc Kim, Lien Huynh Mai (2025), “Financial Characteristics and Sector Context in Classifying Financial Report Fraud: Evidence from Vietnam,” *Asian Journal of Accounting Research* (Scopus Q2, ABDC C).

JOURNAL

1. **Nguyen Luong Gia Bao**, Ha Quoc Hung, Vo Pham Anh Khoa, Ngo Phu Thanh (2025), “Impact of Margin Cuts on Stock Returns and Trading Value: Evidence from the Vietnamese Stock Market,” *Journal of Economics and Finance* (ESCI Q2, IF: 1.5, ABDC B). DOI: 10.1007/s12197-026-09773-x.
2. **Nguyen Luong Gia Bao**, Ngo Phu Thanh (2026), “World Uncertainty and the Marginal Value of Cash: Suggestive Non-linear Evidence from Six ASEAN Technology Markets,” *International Journal of Economics and Business Research* (Scopus Q4, ABDC C) – Accepted. DOI will be updated.

CONFERENCE

1. Ha Quoc Hung, Ngo Phu Thanh, **Nguyen Luong Gia Bao** (2026), “Semantic Stacking for Equity Ranking: Cross-Sectional Temporal Encoders with Learning-to-Rank,” *The 10th Vietnam International Conference in Finance (VICIF 2026)* (Presented Paper).
2. **Nguyen Luong Gia Bao**, Ngo Phu Thanh (2026), “World Uncertainty and the Marginal Value of Cash: Suggestive Non-linear Evidence from Six ASEAN Technology Markets,” *7th Vietnam Economist Annual Meeting 2026 (VEAM 2026)* (Presented Paper).
3. **Nguyen Luong Gia Bao**, Ngo Phu Thanh (2026), “State-Dependent Graph Portfolio Optimization using Selective State Space Models and Regime-Adaptive Graph Learning,” *Conference on Optimization, Modeling, Simulation, and Analytics (COMOSA 2026)* (Presented Paper).
4. Tam Phan Huy, Vu Tran Huu Anh, **Nguyen Luong Gia Bao**, Tien Vo Quoc, Thu Truong Ngoc Kim, Lien Huynh Mai (2025), “Financial Characteristics and Sector Context in Classifying Financial Report Fraud: Evidence from Vietnam,” *The 2nd Global Conference on Sustainability in Economics, Business and Law* (Presented Paper).
5. Tam Phan Huy, Vu Tran Huu Anh, Minh Nguyen Hoang, Phuc Huynh Hoang, **Nguyen Luong Gia Bao**, Trieu Nguyen Hoang (2025), “Causality Analysis of Economic Policy Uncertainty and Volatility in Clean Energy Stock Indexes,” *University of Economics and Law – Student Research Award*.
6. Tam Phan Huy, Vu Tran Huu Anh, **Nguyen Luong Gia Bao**, Tien Vo Quoc, Thu Truong Ngoc Kim, Lien Huynh Mai (2025), “A Multi-Layer Machine Learning and Reinforcement Learning Framework for Portfolio Optimization in VN-100,” *University of Economics and Law – Student Research Award*.

ACTIVITIES

AI4I Club

November 2023 – February 2025

External Relations Officer → Marketing and Communications Officer

- Led external relations initiatives and strategic partnership development with industry professionals
- Coordinated collaborative projects with academic institutions and technology companies
- Managed comprehensive marketing campaigns and created engaging content for social media platforms

ACADEMIC HONORS

- Honorable Mention in Scientific Research – University of Economics and Law 2026
- Gold level – WorldQuant BRAIN 2025
- Top 10 – Data Science Competition 2025
- Student of 5 Merits – Faculty of Finance and Banking, University of Economics and Law 2024
- Top 12 – Business Intelligence Competition Season 8 2024
- Top 9 – SAP ERPSIM Student Competition 2024

TECHNICAL PROFICIENCIES

- **Programming:** Python, R, SQL, MATLAB, LaTeX
- **ML & AI:** Deep Learning, Time Series Modeling, Reinforcement Learning, Natural Language Processing
- **Quantitative Finance:** Portfolio Optimization, Risk Modeling, Market Microstructure, Financial Modeling
- **Tools:** PyTorch, TensorFlow, Pandas, NumPy, Jupyter Notebook, Git, VBA, Bloomberg Terminal

LANGUAGES

- **Vietnamese:** Native
- **English:** TOEFL iBT 4.5/6.0 (90)

REFERENCES

Ha Quoc Hung, MSc
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RMIT University Vietnam
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